

Domain and Range: All Three Notations

TEACHER KEY | Algebra II | Unit 1 | Day 1 | TEK A2.7.I

Objective: I can write the domain and range of a function from a graph, a table, or a set of ordered pairs using a verbal description, inequality notation, set notation, and interval notation.

Vocabulary

Domain — the set of all **input** values, the **x**-values.

Range — the set of all **output** values, the **y**-values.

Set notation — uses **braces { }** with the bar |, read "such that."

Interval notation — uses **[]** if an endpoint is included, **()** if it is excluded.

Inequality notation — writes domain or range using **<, ≤, >, or ≥** instead of interval or set notation.

Part 1 — From a Set of Ordered Pairs

Least to greatest, and each value written only once.

Example 1. { (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) }

Domain: { **-4, 0, 2, 3, 7** }

Range: { **-7, -6, -1, 4, 7** }

You Try 1. { (-5, 3), (-1, 0), (2, 3), (6, -8) }

Domain: { **-5, -1, 2, 6** }

Range: { **-8, 0, 3** }

Part 2 — From a Table

Example 2.

x	-3	-1	0	2	5
f(x)	9	1	0	4	25

Domain: { **-3, -1, 0, 2, 5** } Range: { **0, 1, 4, 9, 25** }

You Try 2.

x	-2	0	1	4
f(x)	7	7	-3	10

Domain: { **-2, 0, 1, 4** } Range: { **-3, 7, 10** }

Part 3 — The Three Notations

The first row is done for you. Fill in the rest.

Verbal description	Inequality	Set notation	Interval notation
all real numbers	$-\infty < x < \infty$	$\{ x \mid x \in \mathbb{R} \}$	$(-\infty, \infty)$
all reals greater than or equal to 0	$x \geq 0$	$\{ x \mid x \in \mathbb{R}, x \geq 0 \}$	$[0, \infty)$
all reals less than 5	$x < 5$	$\{ x \mid x \in \mathbb{R}, x < 5 \}$	$(-\infty, 5)$

all reals from -3 to 3, including both	$-3 \leq x \leq 3$	$\{x \mid x \in \mathbb{R}, -3 \leq x \leq 3\}$	$[-3, 3]$
just the numbers -2, 0, and 4	(none)	$\{-2, 0, 4\}$	does not apply

Key idea: bracket [] = endpoint **included**. Parenthesis () = endpoint **excluded**. ∞ always takes a **parenthesis**.

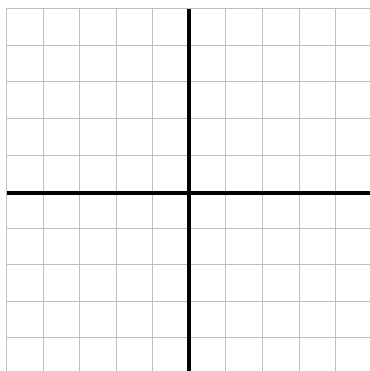
You Try 3. all reals greater than -8 $\rightarrow$ inequality $x > -8$ interval $(-8, \infty)$

You Try 4. $y \leq 12 \rightarrow$ in words **all reals less than or equal to 12** interval $(-\infty, 12]$

Part 4 — From a Graph

Each square is 1 unit. The axes are the two heavy lines, and both grids run -5 to 5.

A. Discrete. Plot $(-4, 2)$, $(-2, -3)$, $(1, 4)$, $(3, -1)$.

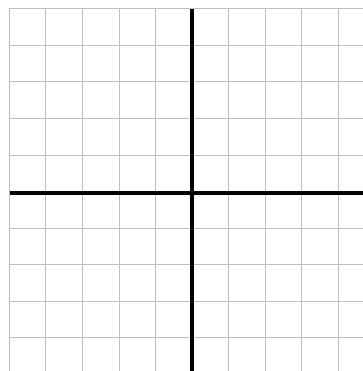


Domain (set): $\{-4, -2, 1, 3\}$

Range (set): $\{-3, -1, 2, 4\}$

Continuous or discrete? **discrete**

B. Continuous. Graph $y = x + 1$ from $x = -4$ to $x = 3$.



Domain (inequality): $-4 \leq x \leq 3$

Domain (interval): $[-4, 3]$

Range (interval): $[-3, 4]$

Part 5 — Continuous or Discrete?

Write continuous or discrete.

1. The number of students in each class period **discrete**
2. The outside temperature over one full day **continuous**
3. The total cost of n concert tickets at \$45 each **discrete**

Exit Ticket

Given $\{(1, 5), (3, 5), (4, -2)\}$ write the domain and range in set notation, then say whether interval notation could be used.

Domain $\{1, 3, 4\}$ Range $\{-2, 5\}$

Could interval notation be used? **no - the relation is discrete**